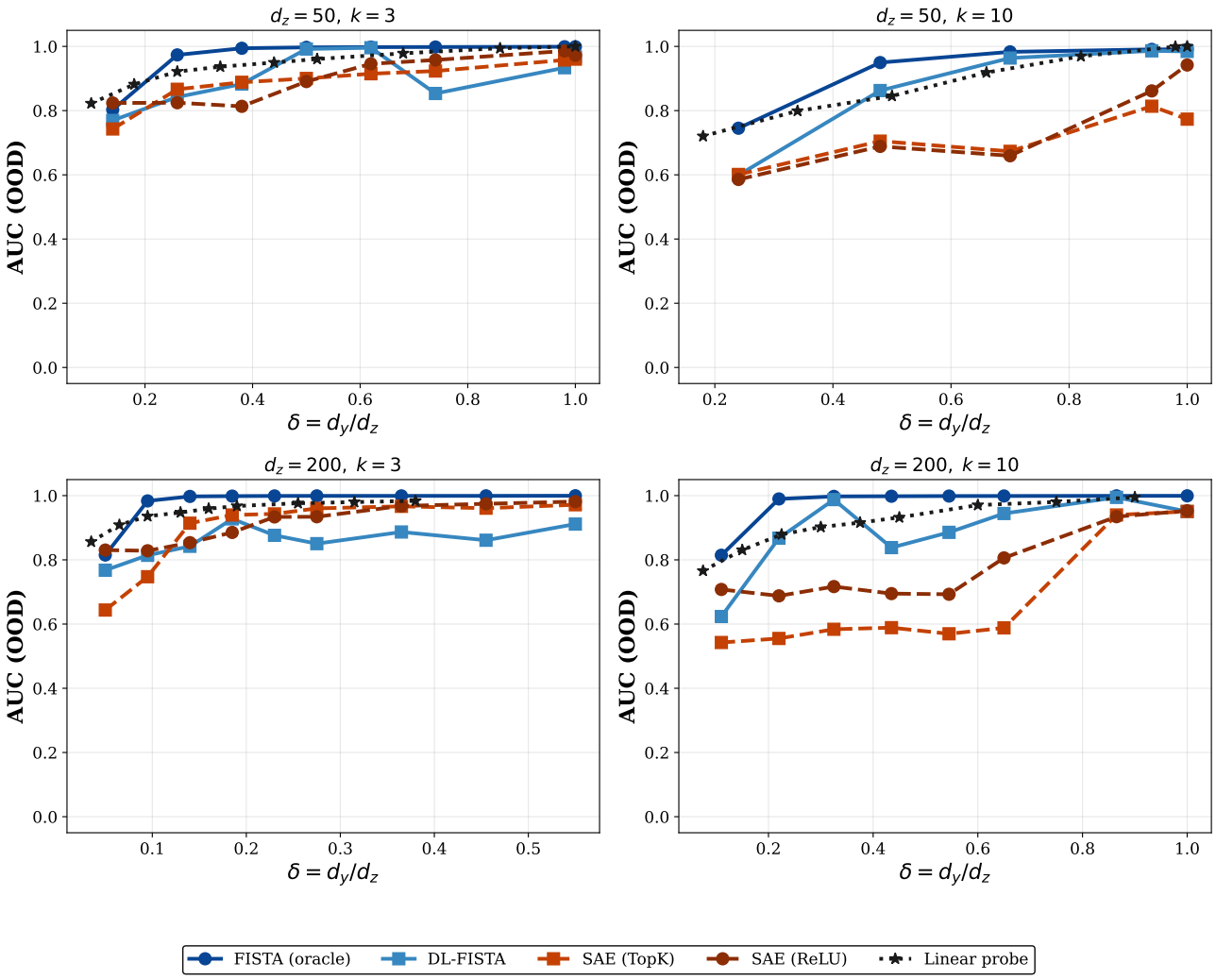
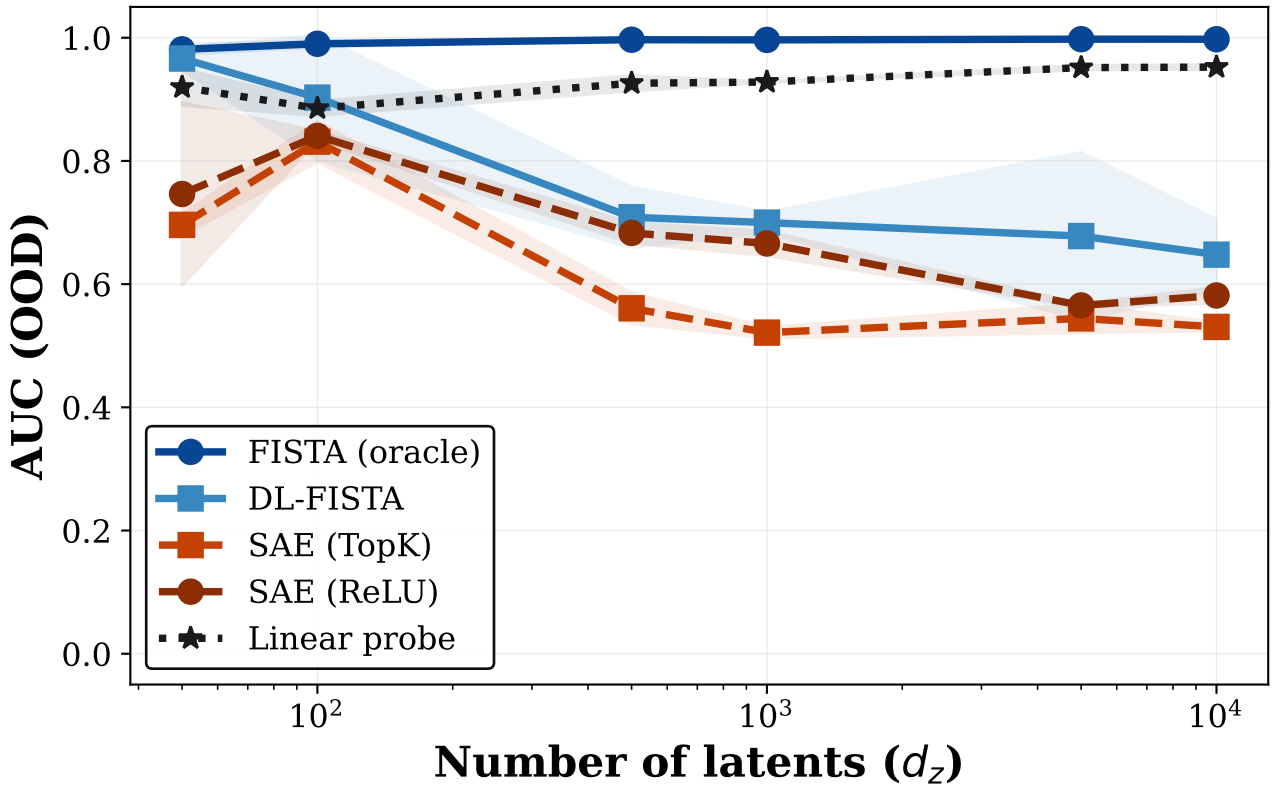


Figure 17: **Phase transition: AUC (OOD).** SAE OOD AUC degrades toward chance while per-sample methods maintain near-perfect performance across all (d_z, k) settings.



SAEs exhibit a persistent ID–OOD gap; FISTA maintains consistent performance.

Figure 18: **Scaling latent dimension: AUC (OOD).** The linear probe maintains high AUC (~0.93) even as its MCC collapses to 0.07 (Figure 6), illustrating how a supervised single-label metric can mask identifiability failure. FISTA (oracle) dominates; SAEs degrade. $k = 10$, $p = 5000$.

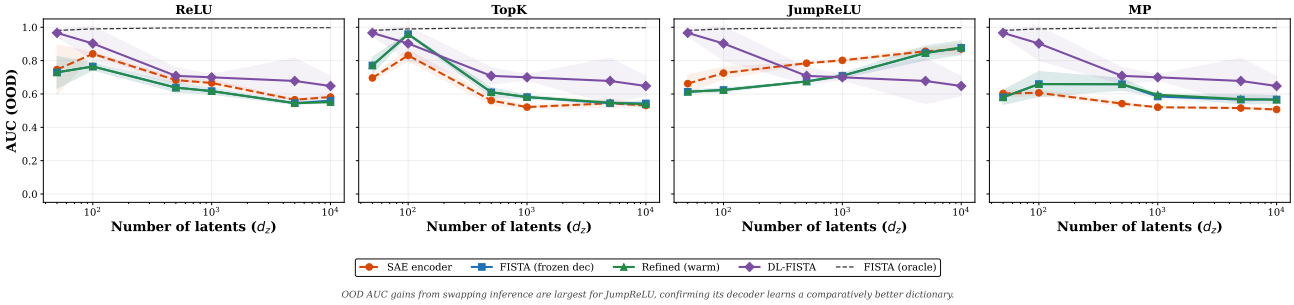
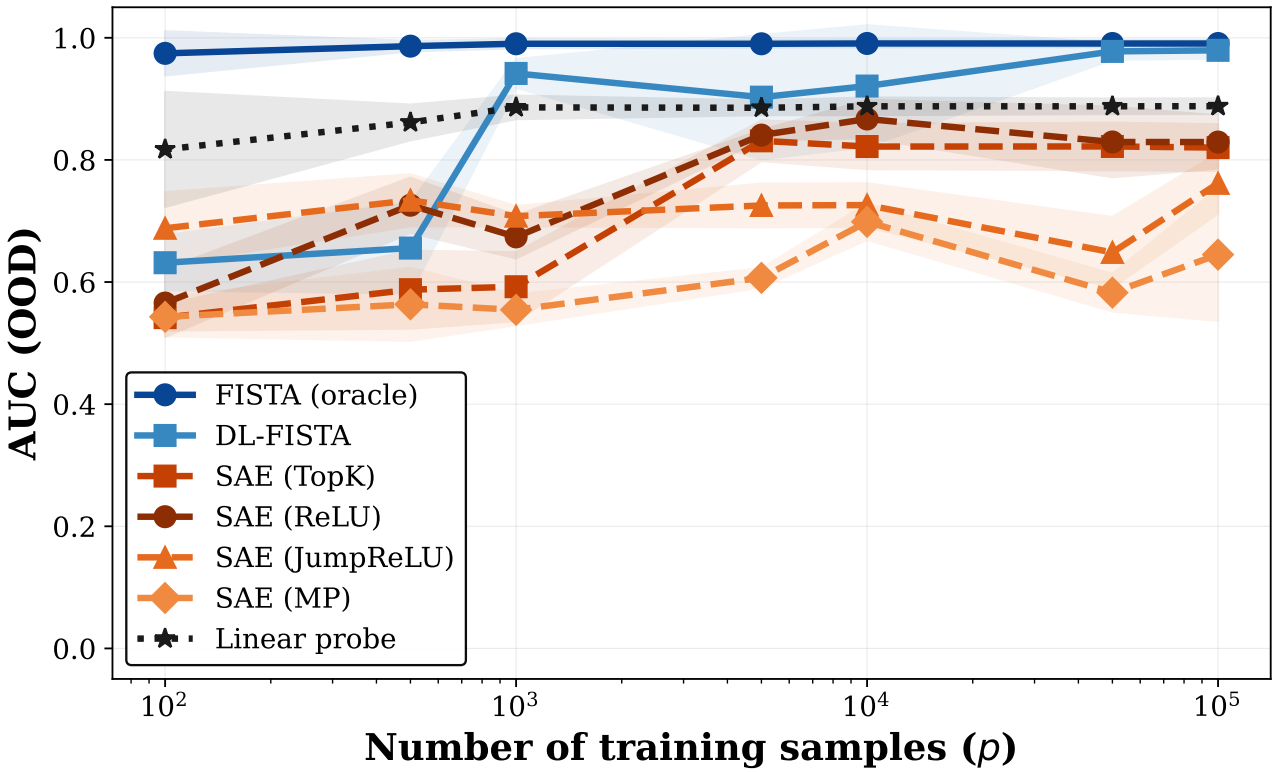


Figure 19: **Frozen decoder ablation: AUC (OOD).** Same layout as Figure 8. FISTA on frozen TopK and JumpReLU decoders yields modest OOD AUC gains, but the gap to DL-FISTA remains large for all types. $k = 10$, $p = 5000$.



Additional training data benefits dictionary quality but does not close the amortisation gap.

Figure 20: **More data: AUC (OOD).** Additional training data benefits DL-FISTA but does not close the amortisation gap on OOD AUC. $d_z = 100$, $k = 10$, $d_y = 47$.